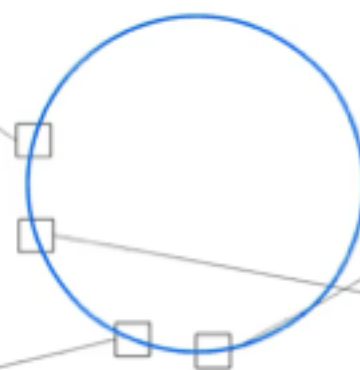
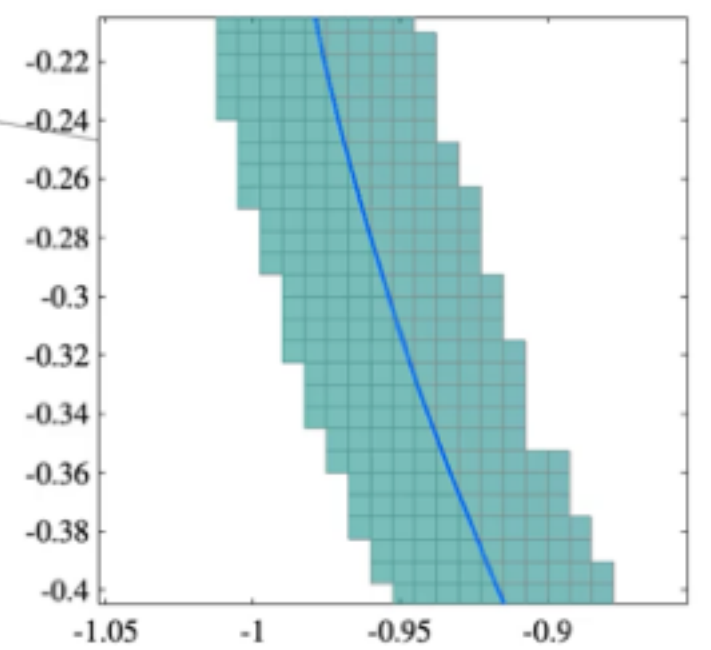
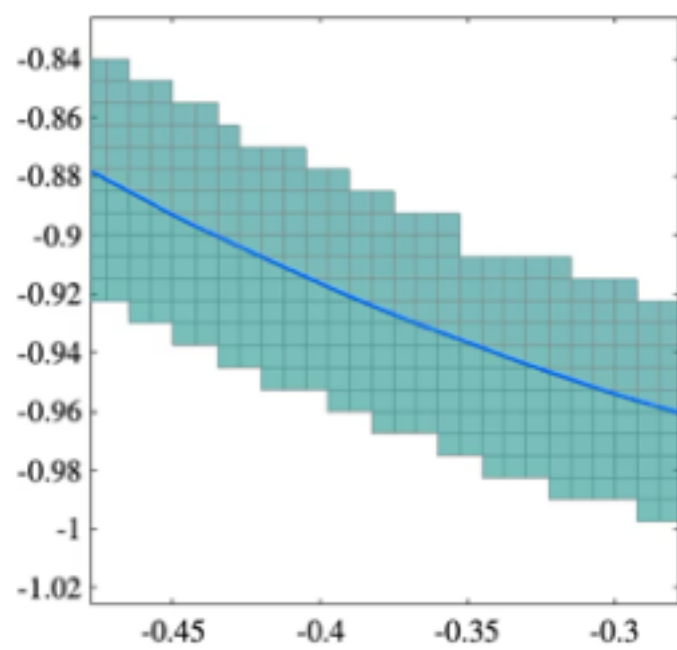
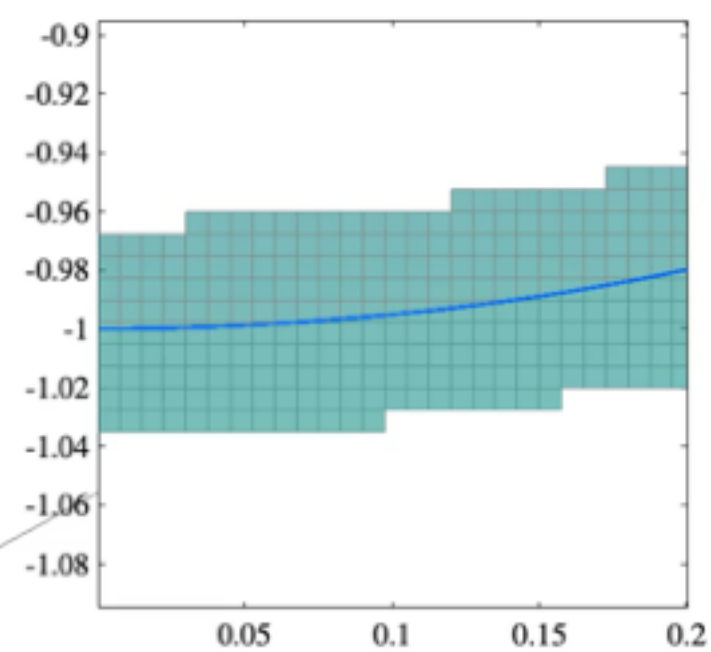
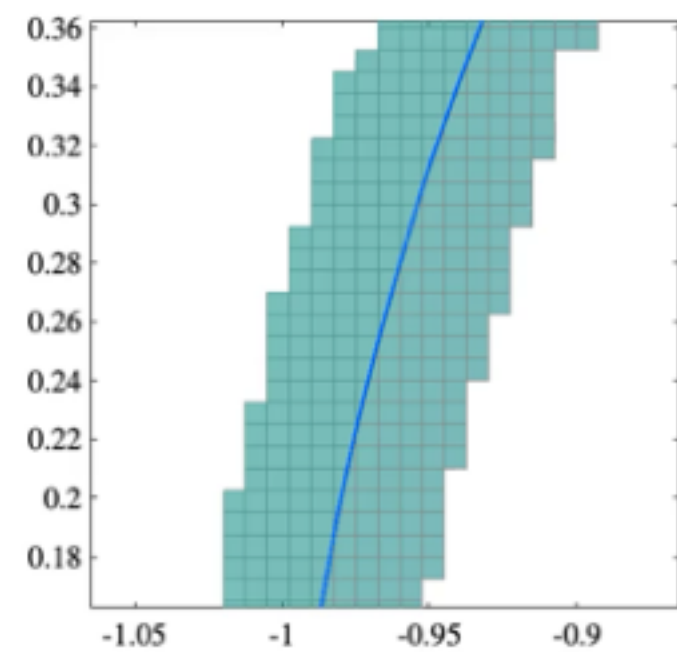


MICROSCALE FILAMENTATION BY ALTERNATING SHEARS



$h/L = 1.25\text{e-}03$, $t = 0$, $N(t) = 1$



TOPOLOGICAL RIGIDITY

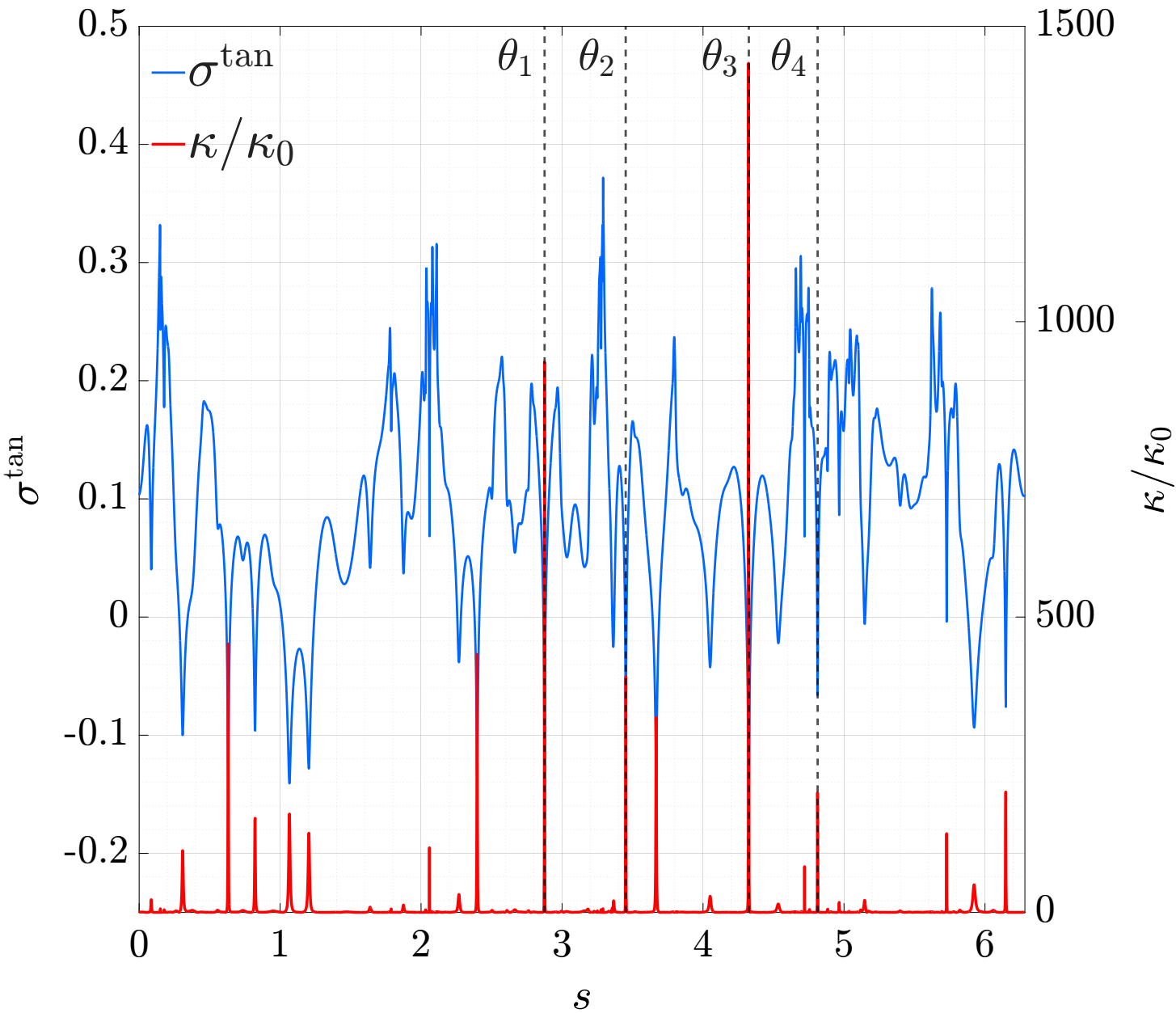
Filaments collapse below the microscale

$$\frac{h}{L} = \mathcal{O}(10^{-3})$$

MULTISCALE CHAOTIC FLOW

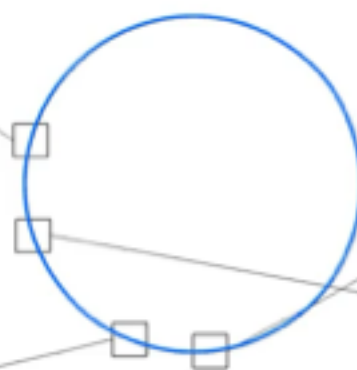
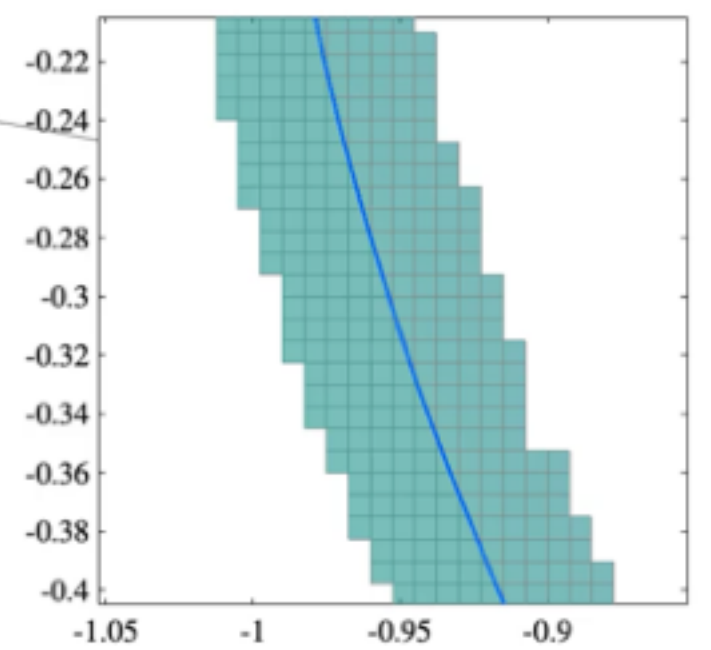
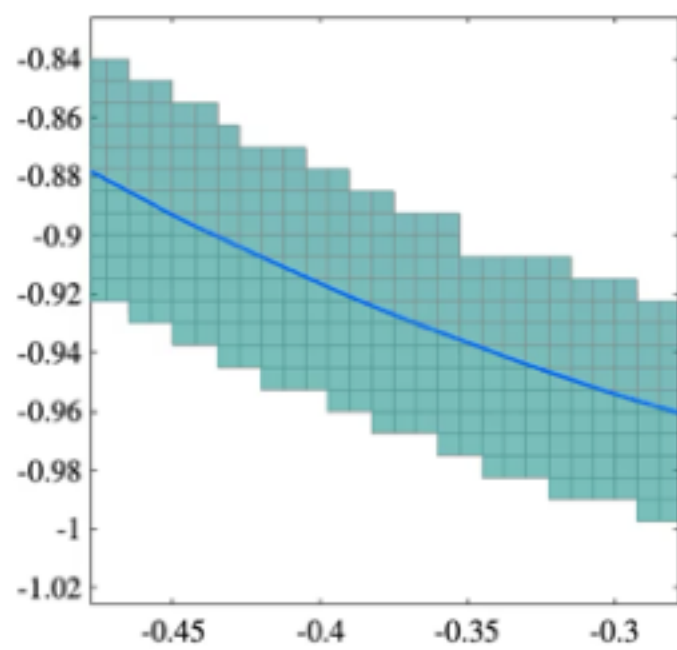
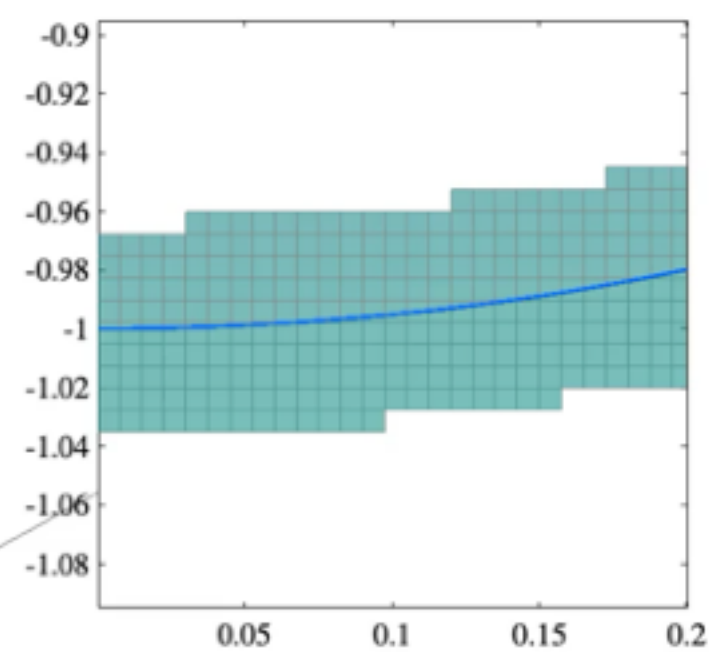
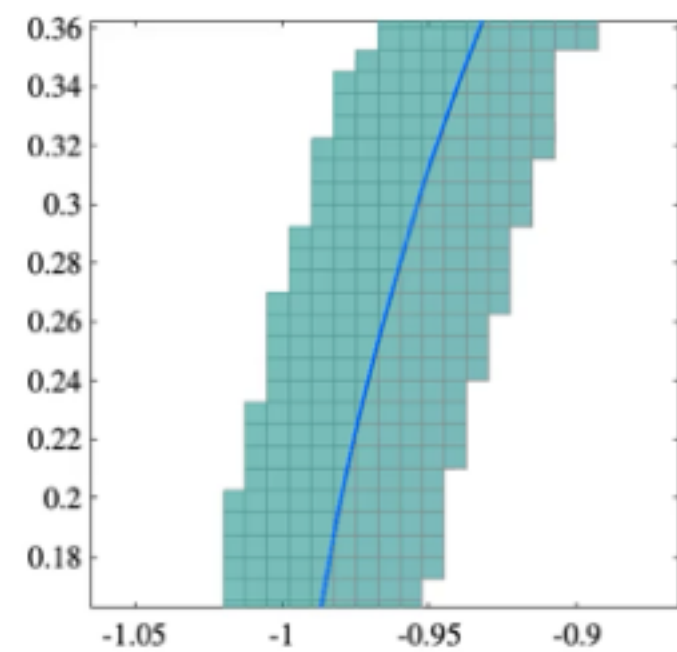
Alternating shears generate exponential stretching and transverse compression.

Geometric & material diagnostics predict where filaments emerge.

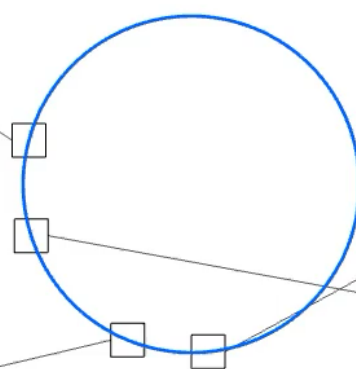
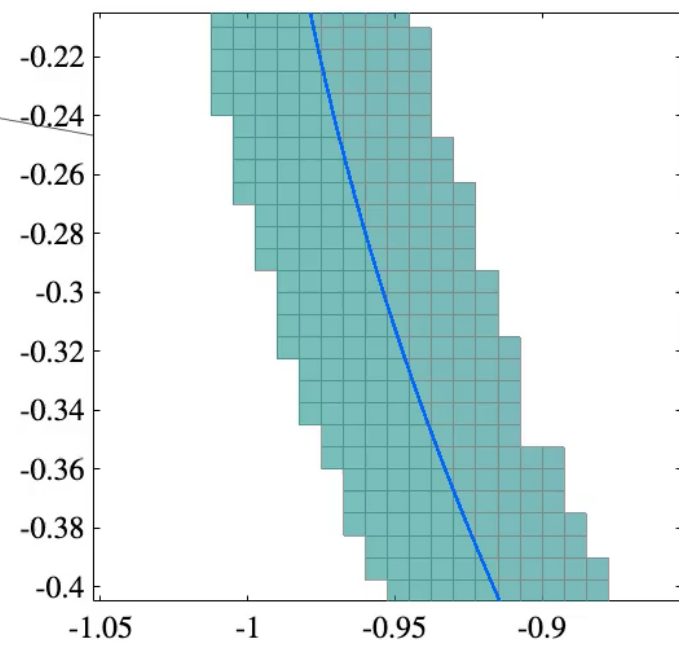
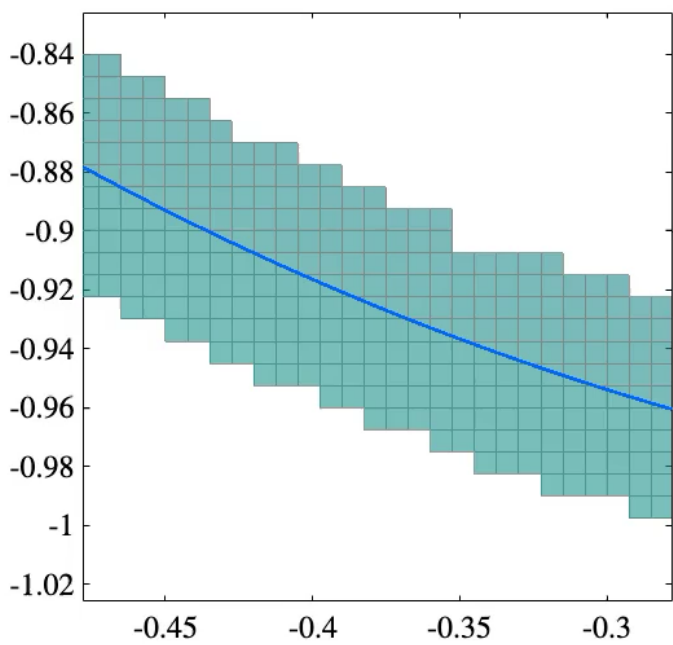
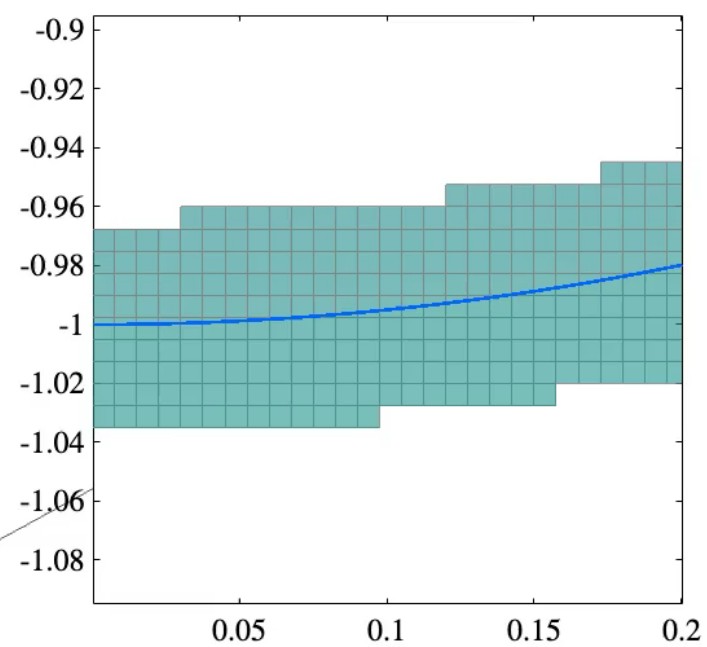
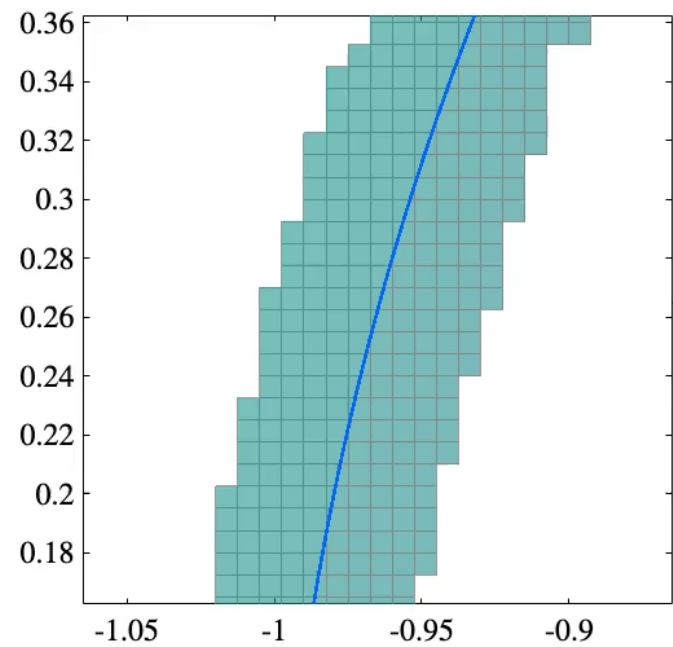




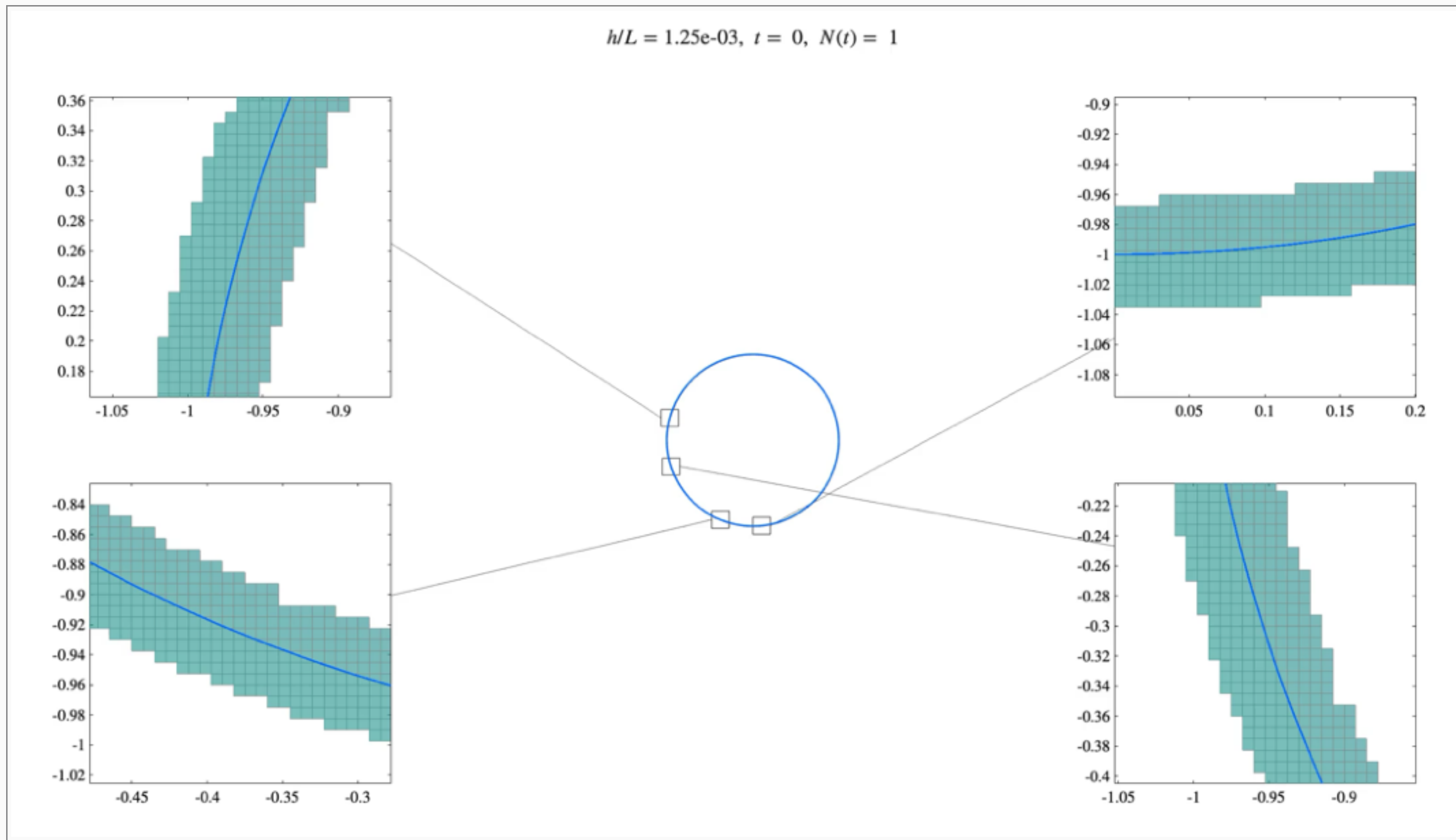
$$h/L = 1.25\text{e-}03, \quad t = 0, \quad N(t) = 1$$



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MICROSCALE FILAMENTATION BY ALTERNATING SHEARS



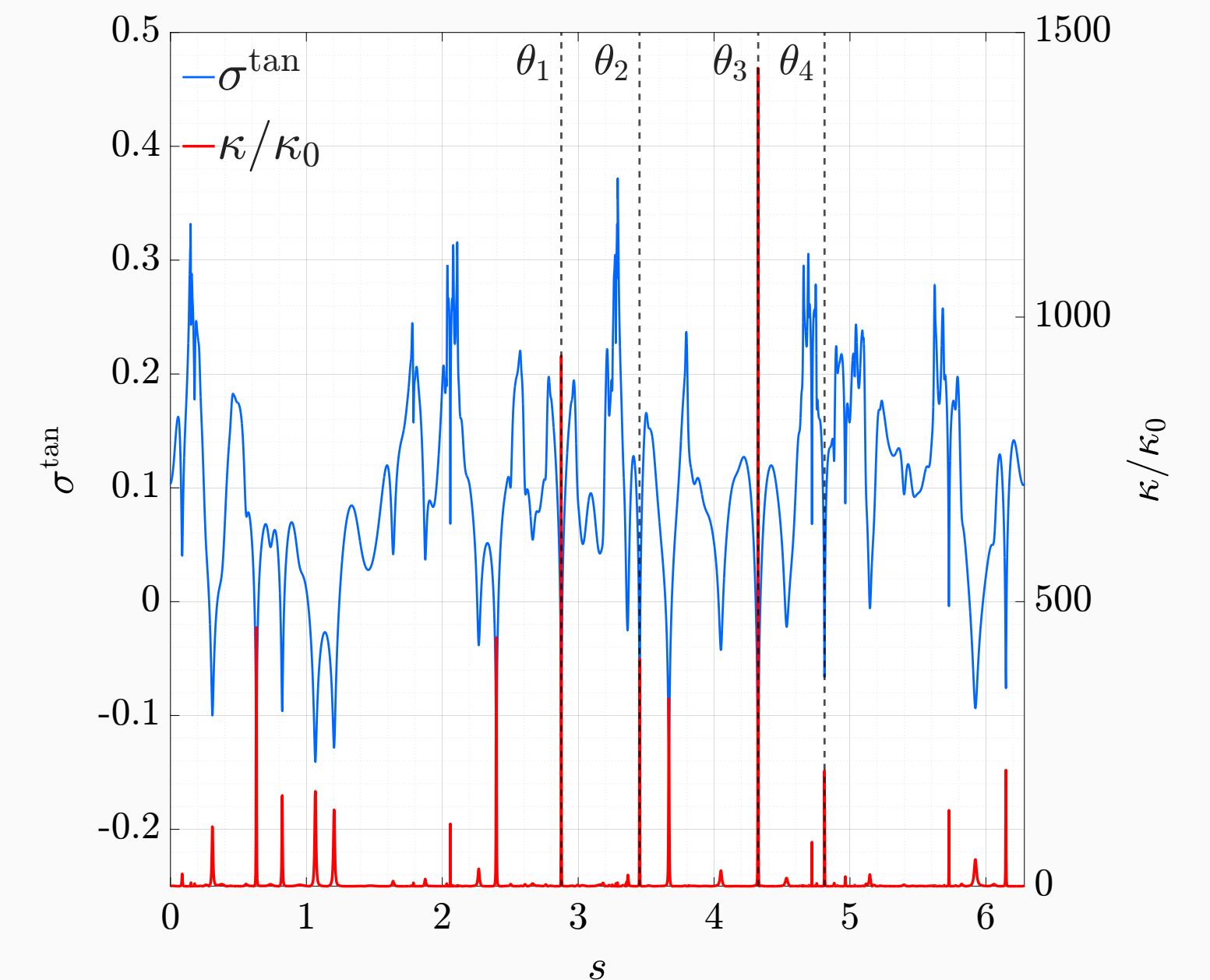
TOPOLOGICAL RIGIDITY

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MULTISCALE CHAOTIC FLOW

Alternating shears generate exponential stretching and transverse compression.

Geometric & material diagnostics predict where filaments emerge.





Quasi-incompressible Cahn–Hilliard fluids and topological transitions

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DIFFUSE-INTERFACE REGULARIZATION

Filament collapse is regularized

by a diffuse layer of width ϵ

